Digital Forensics and Incident Response

Lab: RAM Dump Analysis Using Volatility 3

Memory Dump Download

https://mega.nz/#!6l4BhKIb!l8ATZoliB_ULlvlkESwkPiXAETJEF7p91Gf9CWuQI70

1. Identify Operating System and Kernel Details

Question:
How can you determine the operating system version, kernel build, and system architecture of the given memory dump using Volatility 3?

2. List Active Running Processes

Question:
How can you identify all active processes running on the system at the time the memory image was acquired?

3. Detect Hidden or Terminated Processes

Question:
How can you detect hidden, terminated, or unlinked processes present in the system memory?

4. Analyze Process Hierarchy

Question:
How can you analyze the parent–child relationship of processes to identify suspicious process spawning?

5. Identify Malicious Code Injection

Question:
How can you detect suspicious or malicious memory regions indicating possible code injection in running processes?

6. Perform Network Connection Analysis

Question:
How can you identify active network connections, listening ports, and remote IP addresses from the memory dump?

7. Analyze User Sessions

Question:
How can you identify active and disconnected user sessions present in the memory image?

8. Extract Command Execution History

Question:
How can you extract command-line arguments and console command history executed by users from memory?

9. Extract Registry Hives from Memory

Question:
How can you identify and extract Windows registry hives stored in system memory?

10. Identify Persistence Mechanisms via Registry

Question:
How can you identify malware persistence mechanisms using registry Run keys stored in memory?

Conclusion

This lab demonstrates the use of Volatility 3 to analyze a Windows memory dump and extract critical forensic artifacts such as system information, running processes, network connections, user activity, and persistence mechanisms. These techniques are essential for incident response and memory forensics investigations.